THIS IS NOT A CERTIFIABLE DOMESTIC APPLIANCE

Methanol has no ASHRAE refrigerant designation for this service and grades as both toxic and highly flammable. There is no path to IEC 60335-2-40 approval for indoor use.

HAZARD DATA		
PROPERTY	**VALUE**	**CONSEQUENCE**
Charge	**4.9 kg methanol**	**outdoor only**
Flash point	11 C closed cup	below ambient
Flammable range	6 – 36 % in air	unusually wide
Flame	nearly invisible in daylight	cannot be seen
Acute toxicity	10 mL blinds, 30 mL can kill	skin absorbs it
Occupational limit	200 ppm TWA	low
Operating pressure	8.4 bar at 130 C	vessel, not plumbing
Stagnation pressure	**14.0 bar at 150 C**	**the design case**

DESIGN CONSEQUENCES ALREADY BUILT IN

- All-welded SS 316L and copper envelope. No elastomer, no thread, no service port anywhere wetted.
- One-time permanent charge, sealed by cold-weld pinch-off and a welded cap. The machine is replaced, never opened.
- No pressure relief to atmosphere. The tubular adsorber contains stagnation at 4.9 x margin instead of venting methanol.
- Welded stainless containment tray under the evaporator, receiver and condenser, drained to a bund sized for the whole charge.
- Ice box lined and sealed: a leak cannot reach the ice. The ice is not potable in any case.

MACHINE SUMMARY	
Designed output	5 kg ice / day
Solar COP	0.110
Collector	1.0 m2
Total mass, charged	91 kg
Certification status	**NONE — demonstrator**

REQUIRED MARKING

ON THE VESSEL AND THE FRAME

METHANOL CH3OH

4.9 kg — TOXIC, FLAMMABLE

INVISIBLE FLAME

OUTDOOR USE ONLY · DO NOT OPEN · DO NOT VENT

ICE NOT POTABLE · NO IGNITION SOURCE WITHIN 5 m

COMMISSIONING		
STEP	**COMMISSIONING**	**RECORD**
1	Site: outdoors, level, free air, 5 m from ignition	site sheet
2	Confirm bund and tray drain clear	visual
3	Leak test 2 witnessed before first sun	He, < 1e-8
4	First solar cycle observed end to end	T bed, T receiver
5	Confirm condenser drains fully to the receiver	no hold-up
6	Measure ice after night one	kg, vs 5 kg target
7	Record stagnation temperature on a no-load day	T max

IF THE MACHINE IS DAMAGED

- Do not approach a suspected leak with any ignition source. The flame is invisible in daylight; use a thermal camera or a broom handle.
- Methanol vapour is heavier than air near the source and pools in the bund. Ventilate from upwind.
- Do not attempt to recover the charge. Isolate the machine, let it weather in open air, and dispose of the carbon and vessel through a chemical waste route.
- A machine that has been vented to atmosphere is finished as a machine: air in the bed cannot be pumped out in the field.

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK
TITLE: **SAFETY, MARKING AND COMMISSIONING** OUTDOOR OFF-GRID EQUIPMENT / 4.9 kg METHANOL / NOT A DOMESTIC APPLIANCE			SCALE: **NONE** THIRD ANGLE PROJECTION	UNITS: **mm**
DRAWING No.: **SI-300**			REV: **A**	SHEET: **6 / 6**
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-15